



User Manual

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May 2026

Table of Contents

1	<i>Introduction</i>	2
2	<i>How to get started</i>	2
2.1	Installation guide	2
3	<i>Tutorial</i>	7
3.1	Data Preparation & Upload Guide	7
3.2	Overview of Input and Output of Each Module	7
3.3	FASTAptamer-Preprocess (Optional)	8
3.4	FATSaptamer-Count	9
3.5	FATSaptamer-Recount	11
3.6	FASTAptamer-Translate	11
3.7	FASTAptamer-Motif Search	13
3.8	FASTAptamer-Motif Omit	13
3.9	FASTAptamer-Motif Tracker	14
3.10	FASTAptamer-Motif Discovery	15
3.11	FASTAptamer-Mutation Network	15
3.12	FASTAptamer-Distance	16
3.13	FASTAptamer-Data Merge	17
3.14	FASTAptamer-Sequence Enrichment	18
3.15	FASTAptamer-Position Enrichment	19
3.16	FASTAptamer-Differential Analysis	20
3.17	FASTAptamer-Cluster	21
3.18	FASTAptamer-Cluster Diversity	21
3.19	FASTAptamer-Cluster MSA (Multiple Sequence Alignment)	22
3.20	FASTAptamer-Cluster PHMM (Profile Hidden Markov Model)	23
3.21	FASTAptamer-Recluster	24
3.22	FASTAptamer-Pipeline	26
4	<i>Known issues</i>	27
4.1	SSE connection failure during Preprocess	27

1 Introduction

FASTAptamer 3.0: An End-to-End Platform for Bioinformatic Analysis of Nucleic Acid and Protein Combinatorial Selections.

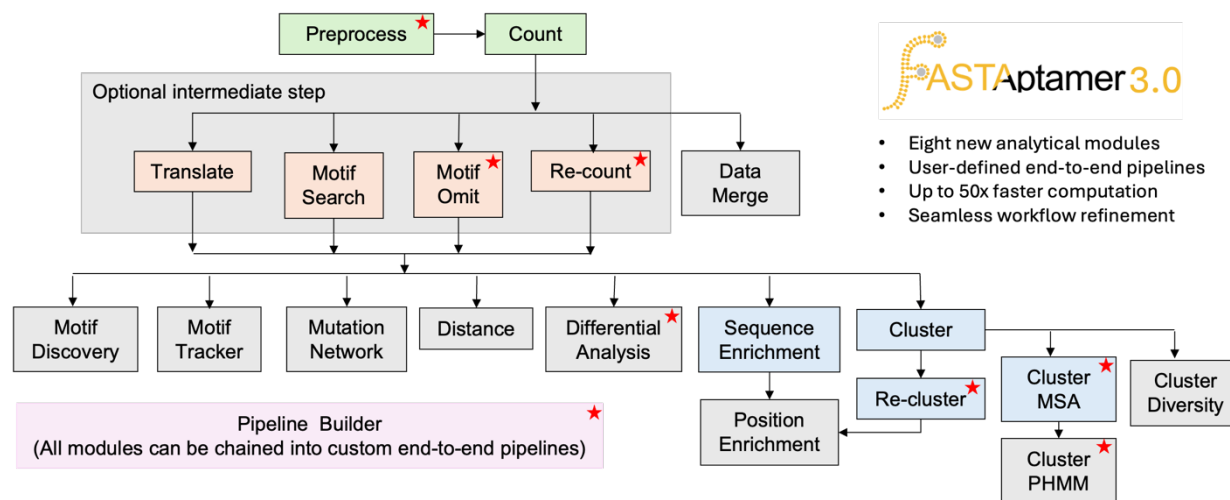


Figure 1. Overview of FASTAptamer 3.0 module connectivity and new features.

Modules new to FASTAptamer 3.0 are indicated by red star icons. Green nodes represent entry-point modules: Preprocess accepts raw FASTQ input; Count serves as the required starting point for all user-defined analytical workflows. Orange and blue nodes represent intermediate analytical steps, with blue nodes such as Cluster MSA feeding into specific downstream analyses. Gray nodes represent terminal outputs. Most modules produce downloadable FASTA, CSV, or both file formats, enabling seamless integration with external bioinformatics tools.

2 How to get started

FASTAptamer 3.0 is an open-source tool built for accessibility and community collaboration within the combinatorial selection field. Users can interact with the software in two ways:

- **Deployment:** Run the user interface locally via [Docker Hub](#) by following the installation guide.
- **Development:** Access the complete source code on [GitHub](#) to contribute to or customize the project.

2.1 Installation guide

2.1.1 Step 1: System Requirements

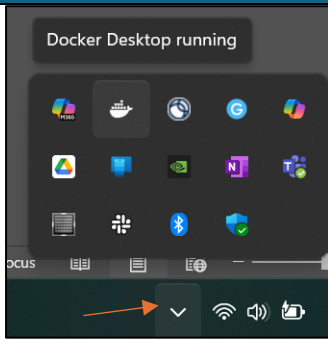
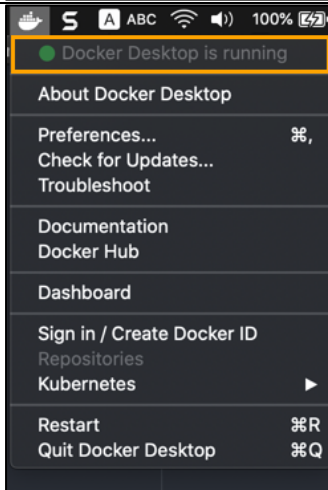
Before starting, make sure your computer meets these requirements:

Component	Windows Requirement	macOS Requirement
Operating System	64-bit Windows 10 or Windows 11	macOS (current and two previous major macOS releases)
User Rights	Administrator privileges required to install Docker Desktop https://docs.docker.com/desktop/setup/install/windows-permission-requirements/	Administrator privileges required to install Docker Desktop https://docs.docker.com/desktop/setup/install/mac-permission-requirements/

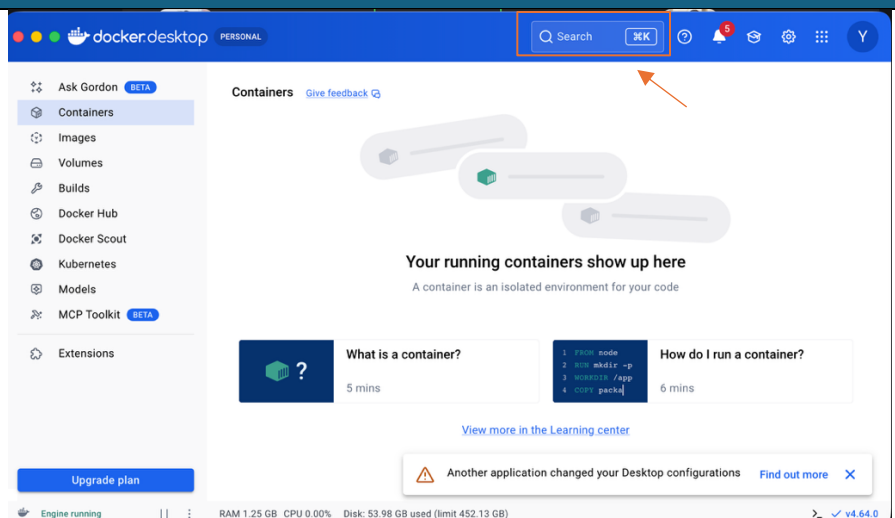
2.1.2 Step 2: Install Docker Desktop

Docker Desktop is required to run the containerized FASTAptamer3 application. Please ensure it is installed and running before proceeding with the FASTAptamer3 installation.

- If Docker Desktop does not start or keeps loading, uninstall it, download a new copy from the Docker website, and reinstall it.
- If you encounter any issues with Docker Desktop, please refer to the official Docker installation documentation.

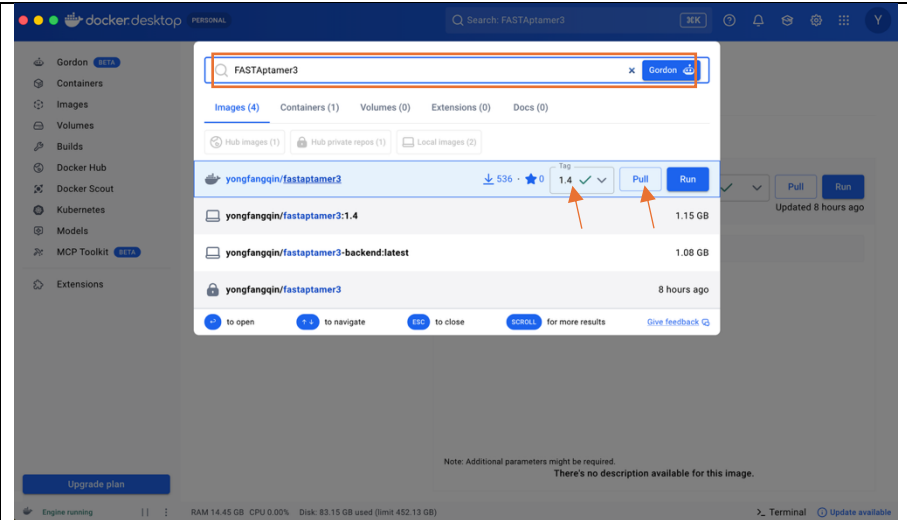
Steps	Screenshot
Windows Users • Download Docker Desktop: https://docs.docker.com/desktop/setup/install/windows-install/ • Run the installer with default settings. If prompted, allow installation of WSL2. Restart your computer if requested. • Open Docker Desktop. • Wait until the whale icon in the Windows taskbar system tray (bottom-right corner of the screen) shows “Docker Desktop is running.”	 A screenshot of the Windows taskbar system tray. A red arrow points to the Docker Desktop icon, which is a whale icon with the text "Docker Desktop is running" next to it.
macOS Users • Download Docker Desktop: https://docs.docker.com/desktop/setup/install/mac-install/ • Choose the correct version: (verify your chip: go to Apple menu → About This Mac) Apple Silicon (M1/M2/M3) → Apple chip version, Intel Mac → Intel version • Double clicks the .dmg file and drag Docker Desktop into the Applications folder. • Launch Docker Desktop. • Wait until the whale icon in the top right menu bar displays “Docker Desktop is running.”	 A screenshot of the macOS menu bar. A red box highlights the Docker Desktop icon, which is a whale icon with the text "Docker Desktop is running" next to it. Below the icon is a menu with options: About Docker Desktop, Preferences..., Check for Updates..., Troubleshoot, Documentation, Docker Hub, Dashboard, Sign in / Create Docker ID, Repositories, Kubernetes, Restart, and Quit Docker Desktop.

2.1.3 Step 3: Download the FASTAptamer3 Image

Steps	Screenshot
1. Open Docker Desktop 2. Click Search	 A screenshot of the Docker Desktop interface. A red box highlights the "Search" button in the top right corner of the interface. A red arrow points to the "Search" button. The interface shows a sidebar with navigation options like Containers, Images, Volumes, Builds, Docker Hub, Docker Scout, Kubernetes, Models, MCP Toolkit, and Extensions. The main area displays "Your running containers show up here" and a "What is a container?" section.

3. Type FASTAptamer3 in the search box.
4. Select the image with the latest version tag.
5. Click Pull to download the image.

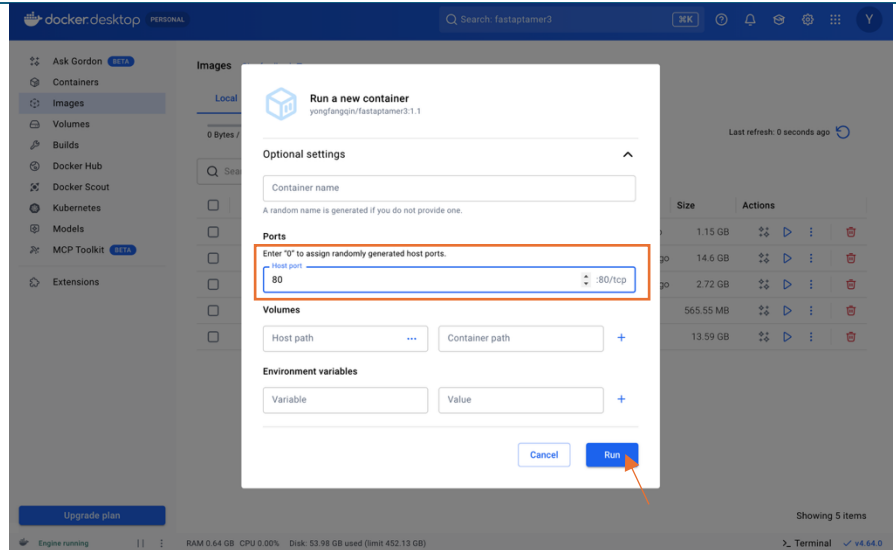
Note: To update FASTAptamer3, repeat step 1-5, pulling the latest version of the image from Docker Hub.



2.1.4 Step 4. Run FASTAptamer3

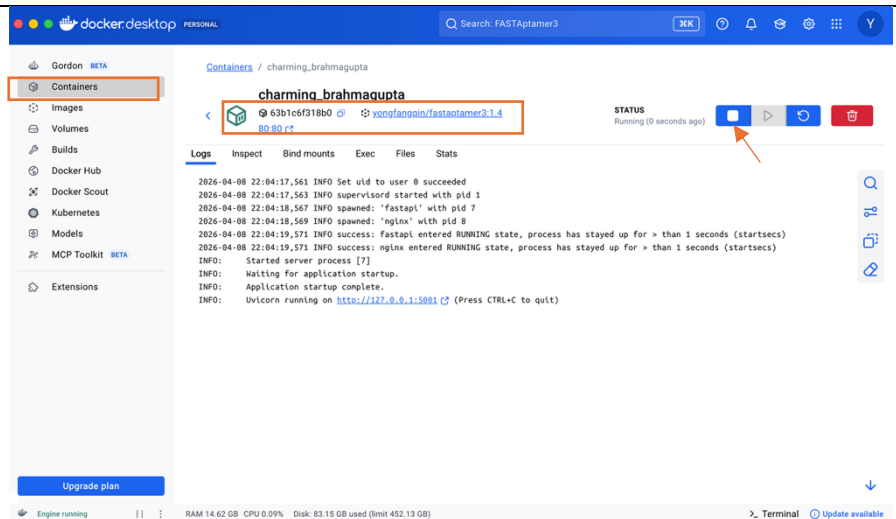
Steps	Screenshot
1. In the Images tab, locate: FASTAptamer3, the tag designation should match the version that you just pulled. (1.4 for example) 2. Under the Actions column, click the Run button.	This screenshot shows the Docker Desktop 'Images' tab. The 'Local' tab is selected, and a search for 'FASTAptamer3' is performed. The results show a list of images. The image 'yongfangqin/fastapter3' with tag '1.4' is highlighted with a red box. A red arrow points to the 'Run' button in the 'Actions' column for this image. The interface also shows other images like 'yongfangqin/mustedeep_web' and 'mongo'.
3. A window will appear asking you to configure the application. 4. Click the Expand down arrow.	This screenshot shows the Docker Desktop interface with the 'Run a new container' dialog box open. The dialog box has a title bar 'Run a new container' and a subtitle 'yongfangqin/fastapter3:1.1'. Below the title bar, there is a section for 'Optional settings' with a downward arrow. A red arrow points to this downward arrow. The dialog box has 'Cancel' and 'Run' buttons at the bottom. The background shows the 'Images' tab with the 'yongfangqin/fastapter3' image highlighted.

5. In the Host Port field, enter any available port number on your computer (e.g., 80, 8080 or 8888). This is the port your browser will use to open the app — for example, entering 80 means you will access the app at `http://localhost:80`. Choose any number between 80 and 65535 that is not already being used by another application on your system. If the selected port is already in use, the application will fail to start and display a warning. Simply return to the Host Port field, enter a different port number
6. Then click Run button.



7. Docker Desktop will automatically navigate to the container logs. If not, you can jump to the container log.
8. The container status should change to Running.
9. This completes installation. FA3 is now ready for you to access.

Important: If you encounter any issues, please copy the logs from Docker Desktop and provide them to us so we can investigate further.



2.1.5 Step 5: Open the FASTAptamer3 Interface

Open your browser and go to <http://localhost:80>. If you used a different port in Step 4.5, replace 80 with your selected port number. You should now see the FASTAptamer3 web interface.

FASTAptamer 3.0

Start Translate Motif Mutation Network Distance Data Merge Enrichment Differential Analysis Cluster Pipeline About

Preprocess Count Recount

Trims constant 5' and 3' region from sequences in a FASTQ or FASTA file, then filters sequences by length range and maximum average error thresholds. Outputs a cleaned FASTA file ready for counting.

Choose data to preprocess *

Browse... FASTQ or FASTA file

*Do not start until step 3 shows 'Ready'.

Constant 5' region:

Constant 3' region:

Adapter matching error rate: 0.1

Min 0 Max 0.5

Fixed bases to trim from 5' end: 0

Processing logs

2.1.6 Step 6: Stopping the Application

Steps	Screenshot
1. Open Docker Desktop. 2. Go to the Containers tab. 3. Click Stop next to the FASTAptamer3 container. 4. Or remove the container by clicking trash button	

3 Tutorial

3.1 Data Preparation & Upload Guide

FASTAptamer-3 utilizes versatile string-based functions to analyze any biological sequence population, including nucleotide or amino acid libraries. To ensure compatibility, please follow the workflow below.

3.1.1 Preprocessing Workflow

Before analysis, data must move through a specific pipeline to be correctly formatted.

Step A: Preprocess (Optional)

If your raw data contains adapters, primers, or low-quality reads, use the Preprocess module. This allows you to:

- Trim specific constant sequences or primers.
- Filter reads based on minimum/maximum length or quality scores.
- Clean your library to improve the accuracy of downstream clustering and enrichment.

Step B: Count (Required)

All libraries must be processed through Count (or Preprocess → Count) before use in any other module. This step generates a Ranked FASTA file by assigning each unique sequence a:

- Read Count: Total abundance.
- Global Rank: Popularity relative to the library.
- RPU (Reads Per Unit): Normalized frequency.

Note: The Ranked FASTA is the standard input expected by Cluster, Motif, Translate, Sequence Enrichment, and all other analysis modules.

3.1.2 How To Upload Your Data into Analytical Modules

1. Select module from tabs across the top to navigate to your desired analysis tool.
2. Upload files to analyze by clicking Browser to select files from your local filesystem.
3. Validation: Ensure the input file has been processed through the Count step if the module requires a "Ranked" or "Counted" FASTA.

Troubleshooting Tip: If you encounter an error, verify that you are not uploading raw FASTQ data to a module that expects a Ranked FASTA. Common issues include non-standard headers or skipping the mandatory Count step.

3.1.3 Sample Data

We provide sample datasets on the FASTAptamer 3.0 Count page. Use these to explore module functionality and verify expected file formats before uploading your own experimental data.

3.2 Overview of Input and Output of Each Module

Table 1. Module Input and Output File Types

Module	Input Files	Output Files
Preprocess	One FASTQ, FASTQ.gz, FASTA or FASTA.gz File	Cleaned FASTA File Ready for Counting
Count	One FASTQ, FASTQ.gz, FASTA or FASTA.gz File	FASTA Or CSV File (Note: FASTA Output From This Module Is Referred to as "Counted FASTA" Files Throughout This Tutorial)
Recount	2 To 5 Counted FASTA Files	FASTA Or CSV File

Translate	One Counted FASTA File	FASTA Or CSV File
Motif Search	One Counted FASTA or CSV File	FASTA Or CSV File
Motif Omit	One Counted FASTA or CSV File	FASTA or CSV File
Motif Tracker	One Counted FASTA or CSV File	Tracker Summary CSV; Enrichment CSV
Motif Discovery	One Counted FASTA or CSV File	CSV File
Mutation Network	One Counted FASTA or CSV File	CSV File
Distance	One Counted FASTA or CSV File, Plus A Query Sequence	FASTA or CSV File
Data Merge	2 To 26 Counted FASTA Files	CSV File
Sequence Enrichment	Two Counted or Clustered FASTA Or CSV Files	CSV File
Position Enrichment	One Recluster CSV, Sequence Enrichment CSV, or Any CSV With an Enrichment Column	CSV File
Differential Analysis	One or More Counted FASTA Files per Condition (2 Conditions)	CSV File
Cluster	One Counted FASTA File	A FASTA or CSV File Containing Sequences Grouped Across All Clusters or A Single-Cluster FASTA File Extracted From Any One Cluster (Note: FASTA Output From This Module Is Referred to as "Clustered FASTA" Files Throughout This Tutorial)
Cluster Diversity	One Clustered FASTA or CSV File	CSV File
Cluster MSA	One Clustered FASTA File	FASTA Or CSV File
Cluster PHMM	One MSA FASTA File (From Cluster MSA)	PHMM Model File; FASTA of Simulated Sequences
Recluster	Two Clustered FASTA or CSV Files (or Three Files In Multi-Round Mode)	Summary CSV; Reclustered Sequences FASTA (2-Population Mode Only)
Pipeline	One FASTQ or FASTA File	Intermediate And Final Result Files From Each Step in the Configured Workflow (Preprocess → Count → Cluster), Downloadable as Each Step Completes

3.3 FASTAptamer-Preprocess (Optional)

3.3.1 Description

Prior to sequence counting and population-level analysis, raw sequencing reads must be processed to remove flanking regions, filter by length, and exclude low-quality reads. In FASTAptamer 3.0, these tasks are handled by the Preprocess module, which is new to the FASTAptamer series and replaces the external preprocessing tools required in prior versions. Preprocess accepts raw FASTQ files as input and performs five sequential operations. First, it identifies and trims the user-specified 5' and 3' flanking regions from each read using a configurable adapter matching error rate, which controls the maximum mismatch tolerance during adapter detection and accommodates minor sequencing errors in the flanking region without discarding valid reads. Second, it applies fixed trimming of a user-specified number of bases from the 5' and/or 3' ends of each read, enabling removal of any remaining primer-derived nucleotides or library-specific overhangs that fall outside the randomized region after adapter trimming. Third, it applies a length filter to remove reads whose trimmed sequence falls outside a user-defined range, accommodating natural variation in read length while excluding truncated or aberrant sequences. Fourth, it calculates the average per-base error rate from the Phred quality scores and removes reads that exceed the user-specified maximum. The output is a FASTA file of quality-filtered, trimmed sequences ready for input into the Count module.

3.3.2 Usage

- Click **Browse...** to upload a FASTQ or FASTA file (.fasta, .fastq, .fa, .fq).
- Configure 5' and 3' primer/flanking sequences for trimming and set their respective adapter matching error rates
- Use the sliders to set the number of fixed bases to trim from the 5' and 3' ends, the minimum and maximum accepted sequence length after adapter trimming, and the maximum allowed error rate.
- Click **Start**. Progress logs appear on the right panel in real time.
- Click **Download** to save the cleaned FASTA output.

3.3.3 Parameters

Parameter	Default	Description
Constant 5' region	-	Enter the flanking sequence at the 5' end to trim (e.g., your primer sequence). All bases before the matched 5' flanking sequence will be removed. Leave blank to skip.
Constant 3' region	-	Enter the flanking sequence at the 3' end to trim. All bases after the matched 3' flanking sequence will be removed. Leave blank to skip.
Adapter matching error rate	0.1	How strictly the adapter sequence must match. A value of 0.1 means up to 10% mismatches/indels are tolerated (e.g. 2 errors for a 20nt adapter). Increase if trimming is too aggressive; decrease for stricter matching (range: 0 to 0.5)
Fixed bases to trim from 5' end	0	Remove a fixed number of bases from the beginning of each read, regardless of sequence content.
Fixed bases to trim from 3' end	0	Remove a fixed number of bases from the end of each read, regardless of sequence content.
Sequence length range (Min / Max)	10-100	Keep only sequences within this length range after trimming. Reads outside this window are discarded (1–500 nt).
Maximum allowed average error probability	0.05	Quality filter based on Phred scores. Lower values = stricter quality. At 0.05, sequences with more than 5% average error probability are discarded (range: 0.001–1.0).

3.4 FATSaptamer-Count (Required)

3.4.1 Description

Upload your FASTA or FASTQ file to count how many times each unique sequence appears in your data. FATSaptamer 3.0 automatically ranks sequences from most to least abundant and calculates a normalized abundance score (Reads Per Unit, RPU) so you can fairly compare results across experiments with different sequencing depths. The output is a ranked sequence table that feeds directly into all downstream analysis steps.

3.4.2 Usage

- Click **Browse...** Upload a FASTA or FASTQ file. Compressed files with .fasta.gz, .fastq.gz, .fa.gz, or .fq.gz extensions are also supported. The input file can be the output from the Preprocessing module, another preprocessing tool, or a previously generated Count result
- If uploading a previously generated count result, click **Visualize Results** to load it directly into the table and plots without re-running.
- Set normalization scale, reverse complement option, and download format.
- Click **Start**. Total and unique sequence counts are displayed after completion.
- Click **Download** to save the output.

3.4.3 Parameters

Parameter	Default	Description
Normalization value (RPU scale)	1e+6	The denominator used to normalize read counts. Options: 1, 10, 100, 1,000, 10,000, 1×10^5 , 1×10^6 . Select to match your downstream analysis convention.
Return reverse complement?	No	If “Yes”, reports the reverse complement of each sequence instead of the original strand. Use if your sequences were read in the opposite direction.
FASTA or CSV download?	FASTA	Choose FASTA output for use in downstream FASTAptamerR modules, or CSV for easy import into spreadsheets and other analysis tools.

3.4.4 Output Table Columns

Column	Description
ID	Label for each sequence (e.g., Rank=1;Reads=5042;RPU=1234)
Rank	Position by abundance; 1 = most abundant
Reads	Raw number of times this sequence was observed
RPU	Abundance normalized to your selected scale
Length	Sequence length
Sequence	Sequence

3.4.5 Plotting

We have Reads per Rank Plot, Sequence-Length Histogram and Abundance Plot in Count module. All plots share two pre-filter sliders:

Parameter	Default	Range	Description
Min. number of reads to plot	10	0–1,000	Exclude sequences with fewer reads from the plot
Max. rank to plot	100	10–1,000	Limit the plot to sequences up to this rank

3.4.6 Reads per Rank Plot

Line chart of total reads per unique sequence (y-axis) vs. rank (x-axis). Shows the abundance distribution of the sequence.

- Click **Reads per rank** to generate.
- Enable **Adjust default reads-per-rank plot?** to customize the plot. You can modify the X-axis label, Y-axis label, plot title, and line color.

3.4.7 Sequence Length Histogram

Stacked bar chart showing the number of unique sequences (top) and total reads (bottom) per sequence length bin.

- Click **Sequence length histogram** to generate.
- Set **Adjust default sequence-length histogram?** To Customization options: X-axis label, Y-axis 1 label (unique), Y-axis 2 label (total), plot title, bar fill colors for unique and total.

3.4.8 Abundance Plot

Bar chart grouping sequences by read-count bins, showing what fraction of the population falls in each abundance tier. A gradient color encodes the number of unique sequences per bin.

- Click **Abundance plot** to generate.

- Set **Adjust default abundance plot?** To customization options: singleton bin toggle, comma-separated breakpoints, X-axis label, Y-axis label, plot title, gradient start/end colors.

3.5 FATSaptamer-Recount

3.5.1 Description

Upload 2 to 5 counted FASTA files to merge them into a single ranked table. FASTaptamer 3.0 consolidates all unique sequences, sums their read counts across files, and recalculates RPU values on your chosen scale. This is useful for combining technical replicates or multiple sequencing runs before downstream analysis.

3.5.2 Usage

- Click **Browse...** and select 2–5 FASTA files at once (or upload them one at a time).
- After all files appear in the upload list, set normalization scale and download format.
- Click **Start**. Total and unique sequence counts are shown after completion.
- Click **Download** to save the recounted output.
- **Note:** Files must all be counted FASTA files (output from Count or Preprocess → Count).

3.5.3 Parameters

Parameter	Default	Description
Normalization value (RPU scale)	1e+6	The denominator used to normalize read counts. Options: 1, 10, 100, 1,000, 10,000, 1×10^5 , 1×10^6 . Select to match your downstream analysis convention.
FASTA or CSV download?	FASTA	Output format.

3.5.4 Reads per Rank Plot

Line chart of total reads per unique sequence (y-axis) vs. rank (x-axis). Shows the abundance distribution of the sequence.

- Click **Reads per rank** to generate.
- Enable **Adjust default reads-per-rank plot?** to customize the plot. You can modify the X-axis label, Y-axis label, plot title, and line color.

3.5.5 Sequence Length Histogram

Stacked bar chart showing the number of unique sequences (top) and total reads (bottom) per sequence length bin.

- Click **Sequence length histogram** to generate.
- Set **Adjust default sequence-length histogram?** To Customization options: X-axis label, Y-axis 1 label (unique), Y-axis 2 label (total), plot title, bar fill colors for unique and total.'

3.6 FASTaptamer-Translate

3.6.1 Description

Upload a counted FASTA file to translate DNA sequences into amino acid sequences using a standard genetic code or a custom codon table. Identical translation products can be automatically merged and re-ranked by total read count, producing a single ranked table of unique amino acid sequences for downstream analysis.

3.6.2 Usage

- Click **Browse...** to upload a counted FASTA file.
- If uploading a previously generated Translate result file, click **Visualize Results** to load it directly into the table or plots without re-running.
- To load an existing result, click **Visualize Results**.

- Select open reading frame, merge option, and genetic code.
- To use a non-standard codon table, enable **Custom codon table** and enter codon–amino acid pairs (e.g., TGA → W).
- Choose download format.
- Click **Start**. Translated sequences appear in the table.
- Click **Download** to save the output.

3.6.3 Parameters

Parameter	Default	Description
Open reading frame	1	Reading frame offset: 1, 2, or 3. Determines which base to start translation from.
Should non-unique sequences be merged?	Yes	If Yes, identical translation products from different DNA sequences are combined; their read counts are summed, and the sequence is re-ranked.
Genetic code selection	Standard	Dropdown of standard NCBI genetic codes (e.g., Standard, Vertebrate Mitochondrial, Bacterial).
Customize translation table?	No	If Yes, enter comma-separated codons and corresponding amino acid outputs to override the selected code.
Codons to modify	—	Comma-separated codon strings (e.g., AAA, TTT). Active only when customization is enabled.
Custom translations	—	Comma-separated single-letter amino acid outputs, in the same order as the codons above (e.g., Z, X).
FASTA or CSV download?	FASTA	Output format.

3.6.4 Plotting

We have Reads per Rank Plot, Sequence-Length Histogram in Translate module. All plots share two pre-filter sliders:

Parameter	Default	Range	Description
Min. number of reads to plot	10	0–1,000	Exclude sequences with fewer reads from the plot
Max. rank to plot	100	10–1,000	Limit the plot to sequences up to this rank

3.6.5 Reads per Rank Plot

Line chart of total reads per unique sequence (y-axis) vs. rank (x-axis). Shows the abundance distribution of the sequence.

- Click **Reads per rank** to generate.
- Enable **Adjust default reads-per-rank plot?** to customize the plot. You can modify the X-axis label, Y-axis label, plot title, and line color.

3.6.6 Sequence Length Histogram

Stacked bar chart showing the number of unique sequences (top) and total reads (bottom) per sequence length bin.

- Click **Sequence length histogram** to generate.
- Set **Adjust default sequence-length histogram?** To Customization options: X-axis label, Y-axis 1 label (unique), Y-axis 2 label (total), plot title, bar fill colors for unique and total.

3.7 FASTAptamer-Motif Search

3.7.1 Description

Search your counted FASTA file for sequences containing one or more motifs (nucleotide, amino acid, or any string pattern). When multiple motifs are entered, enable Return partial matches to return sequences matching any one motif, or disable to require all motifs to be present.

3.7.2 Usage

- Click **Browse...** to upload a counted FASTA file.
- Enter one or more comma-separated patterns in the Comma-separated patterns field (e.g., AAA,GTG,UUCG).
- Set pattern type, partial match behavior, and optional parenthesization to the output file.
- Set max mismatches allowed (0 = exact match).
- Choose download format (FASTA or CSV).
- Click **Start**. Matching sequences appear in the table.
- Click **Download** to save the output.

3.7.3 Parameters

Parameter	Default	Description
Comma-separated patterns	—	One or more motif strings to search for, separated by commas. Supports nucleotide (IUPAC), amino acid, or plain string patterns.
Place patterns in parentheses in output?	No	If Yes, matched motif occurrences in the output sequence string are wrapped in parentheses for visual identification.
If multiple patterns, return partial matches?	No	If Yes (OR logic): return sequences matching at least one pattern. If No (AND logic): return only sequences matching all patterns.
Max mismatches allowed (0 = exact match)	0	This mismatch tolerance is applied to all motif patterns. For example, if you enter 3 patterns and set max mismatches to 1, each pattern is searched with up to 1 mismatch.
Type of pattern?	Nucleotide	Nucleotide / AminoAcid / String — selects the character matching rules applied.
FASTA or CSV download?	FASTA	Choose FASTA or CSV output format.

3.8 FASTAptamer-Motif Omit

3.8.1 Description

Upload a counted FASTA file to remove sequences containing specified motifs. Enter one or more comma-separated patterns and set Partial Match to Yes (OR) to remove sequences matching any one motif, or No (AND) to remove sequences matching all motifs. This is the inverse of the Motif Search module.

3.8.2 Usage

- Click **Browse...** to upload a counted FASTA file.
- Enter comma-separated patterns to exclude.
- Set partial match behavior and pattern type.
- Set max mismatches allowed (0 = exact match).
- Choose download format.
- Click **Start**. Filtered (non-matching) sequences appear in the table.
- Click **Download** to save the output.

3.8.3 Parameters

Parameter	Default	Description
-----------	---------	-------------

Comma-separated patterns	—	Motif strings to use as exclusion filters, separated by commas.
If multiple patterns, return partial matches?	NO	If Yes (OR logic): remove sequences matching at least one pattern. If No (AND logic): remove sequences matching all patterns.
Max mismatches allowed (0 = exact match)	0	This mismatch tolerance is applied to all motif patterns. For example, if you enter 3 patterns and set max mismatches to 1, each pattern is searched with up to 1 mismatch.
Type of pattern?	Nucleotide	Nucleotide / AminoAcid / String.
FASTA or CSV download?	FASTA	Output format.

3.9 FASTAptamer-Motif Tracker

3.9.1 Description

Are your best aptamer candidates getting more abundant across selection rounds? This module tracks how specific sequences or motifs change in frequency across up to multiple FASTA files — each representing a selection round or time point. Simply upload your files in order, enter the sequences or motifs to track, and FASTAptamer 3.0 generates a line plot showing frequency trends across rounds plus a summary table with enrichment values.

3.9.2 Usage

- Click **Browse...** to select multiple counted FASTA files.
- After upload, reorder files using drag-and-drop or arrow buttons. Use checkboxes to exclude specific files.
- Enter motifs or sequences (one per line) in the Motif or Sequence List text area.
- Optionally provide display aliases (one per line) in the Alias List.
- Choose Motif or Sequence search mode.
- If Motif mode, choose pattern type (Nucleotide / AminoAcid / String).
- Click **Start Tracking**. Results appear in both the Tracker Results and Enrichment Results tables.
- Click **Tracker Plot** to visualize frequency trajectories.
- Click **Download Summary** or **Download Enrichment** to save each table.

3.9.3 Parameters

Parameter	Default	Description
File order	—	Drag and reorder files to define the population sequence (e.g., round 1 → round N). Unchecked files are excluded.
Motif or Sequence List	—	One query per line. Motifs are partial-match sub-sequences; sequences are full exact matches.
Alias List	—	Optional friendly names for each query, corresponding to the same line order. Used as labels in the plot and table.
Search for motifs or whole sequences?	Motif	Motif: reports frequency of each sub-sequence. Sequence: reports total RPU for each exact match.
Type of pattern? (Motif mode only)	Nucleotide	Nucleotide / AminoAcid / String.

3.9.4 Tracker Plot

Line chart with populations on the x-axis and frequency (% for motifs, total RPU for sequences) on the y-axis. One line per query.

- Click **Tracker Plot** (requires results in the table).

- Set **Adjust default tracker plot?** Yes to enable Adjust tracker plot to customize the X-axis label, Y-axis label, plot title, and color palette.

3.10 FASTAptamer-Motif Discovery

3.10.1 Description

Wondering which short sequence patterns are driving your selection? This module scans your counted FASTA file and automatically identifies sub-sequences that appear more often than expected by chance. Set your desired motif length range, and FASTAptamer 3.0 will test every possible pattern for enrichment, ranking results by statistical significance (Z-score and p-value). Output includes a ranked motif table and a scatter plot highlighting the most significant patterns.

3.10.2 Usage

- Click **Browse...** to upload a counted FASTA or previously generated motif discovery CSV.
- If uploading a previously generated Motif Discovery result file, click **Visualize Results** to load it directly into the table or plots without re-running.
- Set sequence alphabet, minimum reads to consider, and motif length range.
- Click **Start**. Results appear in the table.
- Click **Download** to save the output.

3.10.3 Parameters

Parameter	Default	Range	Description
Sequence alphabet	Nucleotide	Nucleotide / AminoAcid	Selects the character set for sub-sequence enumeration.
Min. number of reads to consider	10	0–1,000	Sequences with fewer reads are excluded from the analysis.
Motif length range (Min / Max)	5-10	3–15	Range of sub-sequence lengths to enumerate and test.

3.10.4 Motif Discovery Plot

Click Motif Discovery Plot to generate a scatter plot of motif rank vs. $-\log_{10}(\text{p-value})$, colored by motif length.

- Click **Tracker Plot** (requires results in the table).
- Enable **Adjust default plots?** to customize the X-axis label, Y-axis label, legend title, plot title, and color palette (Accent, Dark2, Paired, Pastel1, Pastel2, Set1, Set2).

3.11 FASTAptamer-Mutation Network

3.11.1 Description

Trace the shortest mutational path between two sequences through your FASTA dataset. Enter a start and end sequence — FASTAptamer 3.0 will step through your data, selecting sequences that move closer to the end sequence within your allowed edit distance at each step. The result is an ordered path table showing each mutational step from start to finish. Note: this analysis may take a few moments — please remain on the page until it completes.

3.11.2 Usage

- Click **Browse...** to upload a counted FASTA file.

- Enter the Start sequence (the sequence to begin the path from).
- Enter the End sequence (the target sequence to reach).
- Set the Maximum allowable distance per step.
- Click **Start**. The path is shown in the table when complete.
- Click **Download** to save the output.

3.11.3 Parameters

Parameter	Default	Range	Description
Start sequence	—	—	The source sequence for the mutational path. Must be a valid nucleotide or amino acid sequence string.
End sequence	—	—	The destination sequence for the mutational path.
Maximum allowable distance	1	1–5	Maximum Levenshtein edit distance allowed per step along the path. Higher values allow larger jumps between steps.

3.11.4 Output Table Columns

Column	Description
From_Sequence	The sequence at the start of each mutational step
To_Sequence	The sequence at the end of each mutational step
Transition_Cost	The Levenshtein Edit Distance between the two sequences in that step

3.12 FASTAptamer-Distance

3.12.1 Description

Compare a sequence of interest against every sequence in your counted FASTA file. Enter your query sequence and FASTAptamer 3.0 will calculate the Levenshtein Edit Distance to every other sequence, returning a ranked table from most to least similar. Use this module to find sequences resembling a known candidate or to help choose the right edit distance threshold before running the Cluster module.

3.12.2 Usage

- Click **Browse...** to upload a counted FASTA or previously generated distance CSV.
- If uploading a previously generated Distance result file, click **Visualize Results** to load it directly into the table or plots without re-running.
- Type the query sequence into the text field.
- Choose download format (FASTA or CSV).
- Click **Start**. Results appear ranked by distance in the table.
- Click **Download** to save the output. FASTA output is required for subsequent modules; CSV retains all columns from the data table.

3.12.3 Parameters

Parameter	Default	Description
Query sequence	—	The reference sequence against which all population sequences are compared. Enter as a plain nucleotide or amino acid string.
Download format	CSV	FASTA or CSV.

Output Table Columns

Column	Description
ID	Sequence identifier (Rank=1;Reads=12345;RPU=12345;Distance=2)
Rank	Global rank in the original counted file

Reads	Raw read count
RPU	Reads per unit
Sequence	Nucleotide or amino acid sequence
Distance	Levenshtein edit distance from the query sequence

3.12.4 Distance Histogram

Dual bar chart showing unique sequence count (top) and total sequence count (bottom) binned by LED from the query.

- Click **Distance histogram** (requires results in table).
- Enable **Adjust default distance histogram** to customize the X-axis label, Y-axis 1 label (unique sequence count), Y-axis 2 label (total read count), plot title, and bar outline and fill colors

3.13 FASTAptamer-Data Merge

3.13.1 Description

Merge 2 to 26 counted FASTA files into a single comparison table, with one row per unique sequence and one column per file. Choose from three join modes to control which sequences are included — all sequences, sequences present in any file, or sequences present in all files. The merged CSV is the starting point for cross-round comparison and differential enrichment analysis.

3.13.2 Usage

- Click **Browse...** and select 2–26 counted FASTA files.
- After upload, reorder files by dragging cards or using arrow buttons. Uncheck files to exclude them.
- Select the **join type**: Union, Intersection, or Left.
- Click **Start**. The merged table appears on the right.
- Click **Download** to save the merged CSV.

3.13.3 Parameters

Parameter	Default	Description
File order	–	Order of files determines column order in the output. Drag or use arrows to reorder; uncheck to exclude from the merge.
How should the data be joined?	Union	Union : include all sequences from all files (missing values filled with NA). Intersection : include only sequences present in every file. Left : include only sequences from the first file (left join).

3.13.4 Persistence Plot

Bar chart showing the number of unique sequences found in exactly 1, 2, 3, ... N populations (where N = number of files). Shows sequence persistence across rounds.

- Click **Persistence plot** after running.
- Enable **Adjust default persistence plot** to customize the X-axis label, Y-axis label, plot title, and bar outline and fill colors.

3.13.5 UpSet Plot

Visualizes set intersections between all uploaded populations in an UpSet diagram format. Shows the number of unique sequences shared by specific combinations of populations.

- Click **UpSet plot** after running.
- Enable **Adjust default UpSet plot** to customize the X-axis label, Y-axis label, and bar fill color.

3.14 FASTAptamer-Sequence Enrichment

3.14.1 Description

Compare sequence abundance between two counted FASTA files. FASTAptamer 3.0 calculates the $\log_2$ enrichment (fold change in RPU) for every sequence shared between the two populations, showing which sequences are becoming more or less abundant. The output table ranks sequences by their enrichment score — positive values indicate selection for, negative values indicate selection against.

3.14.2 Usage

- Upload **Input data 1** (earlier population, e.g., pre-selection or earlier round).
- Upload **Input data 2** (later population, e.g., post-selection or later round).
- Choose whether to keep sequences not found in both files.
- Click **Start**. Results appear in the table.
- Click **Download** to save the output CSV.

3.14.3 Parameters

Parameter	Default	Description
Keep missing sequences?	No	If Yes (union): include sequences found in only one population, with NA for missing values. If No (intersection): include only sequences present in both files.

3.14.4 Output Table Columns

Column	Description
Sequence	Amino acid or nucleotide sequence
ID.a	ID of the sequence in population 1
Reads.a	Read count from population 1
RPU.a	Normalized abundance in population 1
ID.b	ID of the sequence in population 2
Reads.b	Read count and RPU from population 2
RPU.b	Normalized abundance in population 2
Enrichment	$\text{RPU.b} / \text{RPU.a}$ — ratio of abundance in population 2 vs population 1
Log2E	$\log_2(\text{RPU.b} / \text{RPU.a})$ — positive values indicate enrichment, negative indicate depletion

3.14.5 $\log_2$ (Enrichment) Histogram

Histogram showing the distribution of $\log_2$ enrichment values across sequences.

- Click **$\log_2$ (enrichment) histogram**.
- Enable **Adjust default enrichment histogram** to customize the X-axis label, Y-axis label, plot title, and bar outline and fill colors.

3.14.6 RPU Scatter Plot

Scatter plot with RPU from population 1 on the x-axis and RPU from population 2 on the y-axis. Each point is a sequence.

- Click **RPU scatter plot**.
- Enable **Adjust default enrichment scatter plot** to customize the X-axis label (population 1 RPU), Y-axis label (population 2 RPU), plot title, and point color.

3.14.7 RA Plot

Scatter plot with mean $\log_2(\text{RPU})$ on the x-axis (A) and $\log_2$ fold change on the y-axis (R). A classical representation for identifying differentially enriched sequences.

- Click **RA plot**.
- Enable **Adjust default RA scatter plot** to customize the X-axis label (mean log₂ RPU), Y-axis label (fold change), plot title, and point color.

3.14.8 Enrichment box plot

Box plot showing the distribution of enrichment values grouped by cluster. Useful for comparing enrichment levels across cluster families.

- Click **Enrichment box plot**.
- Enable **Adjust default enrichment box plot?** to customize the X-axis label (clusters), Y-axis label (enrichment value), plot title, box outline color, and box fill color.

3.15 FASTAptamer-Position Enrichment

3.15.1 Description

Discover which positions within your aptamer sequences are most conserved or most variable. This module aligns sequences within a cluster and calculates the mean enrichment at each position, revealing the nucleotide or amino acid preferences that drive selection. Upload a Recluster CSV, Sequence Enrichment CSV, or any CSV containing Sequence and Enrichment columns. Note: alignment is performed internally and may take a few moments for large clusters — please remain on the page until processing completes.

3.15.2 Usage

- Upload an **Input CSV file** — a Recluster CSV (from the Recluster tab), a Sequence Enrichment CSV, or any compatible CSV with Sequence and Enrichment columns.
- If the file contains a cluster column, a **cluster selector** appears — choose which cluster to analyze. If no cluster column is detected, all sequences in the file are analyzed together.
- Choose Type of sequences: Nucleotide or AminoAcid.
- Click **Start**. Results appear in the download-ready output.
- Click **Download** to save the positional enrichment CSV.

3.15.3 Parameters

Parameter	Default	Description
Type of sequences?	Nucleotide	Nucleotide/AminoAcid

3.15.4 Bar Plot

Bar chart with alignment positions on the x-axis and mean enrichment at each position on the y-axis.

- Click **Position Enrichment** after a successful run.
- Enable **Adjust default bar plot** to customize the X-axis label (alignment position), Y-axis label (mean enrichment), plot title, and bar outline and fill colors.

3.15.5 Heatmap

Heatmap with alignment positions on the x-axis, residue identities on the y-axis, and color intensity encoding mean enrichment at each position-residue combination.

- Click **Heatmap** after a successful run.
- Enable **Adjust default heat plot** to customize the X-axis label (alignment positions), Y-axis label (residue identities), legend title (enrichment color scale), plot title, and fill color palette.

3.16 FASTAptamer-Differential Analysis

3.16.1 Description

Performs differential expression analysis between two experimental conditions, each represented by 2 or more counted FASTA files (replicates). Uses the edgeR statistical framework to compute $\log_2$ fold change (logFC), $\log_2$ counts per million (logCPM), and a Benjamini–Hochberg corrected p-value for each sequence. Identifies sequences that are significantly more or less abundant in one condition vs. another.

3.16.2 Usage

- Upload 1 or more FASTA files for **Condition 1** (e.g., treatment replicates).
- Upload 1 or more FASTA files for **Condition 2** (e.g., control replicates).
- Set one or more cutoff thresholds to define PClass, including absolute $\log_2$ fold change ($|\log_2\text{FC}|$), minimum $\log_2\text{CPM}$, and adjusted p-value (FDR).
- Click **Start**. Results appear in the table.
- Click **Download** to save the output CSV.

3.16.3 Parameters

Parameter	Default	Range	Description
Absolute $\log_2$ fold change ($ \log_2\text{FC} $) cutoff	1	0.5–20	Minimum absolute $\log_2$ fold-change to classify a sequence as significant. Sequences above this threshold are marked Sig.
Minimum logCPM cutoff	0	0-10	Minimum average abundance (logCPM) required to classify a sequence as significant. Set to 0 to disable. Filters out low-abundance sequences where fold changes may be unreliable.
Adjusted p-value (FDR) cutoff:	1	0-1.0	Maximum adjusted p-value (Benjamini-Hochberg FDR) to classify a sequence as significant. Set to 1.0 to disable (p-value not used). Only applies when ≥ 3 replicates per condition are provided.

3.16.4 Output Table Columns

Column	Description
Sequence	Nucleotide or amino acid sequence
logFC	$\log_2$ fold change between condition 2 and condition 1
logCPM	CPM is calculated from the total library size across all samples (both conditions combined). $\log_2$ counts per million — reflects how abundant a sequence is in the overall dataset. A measure of overall abundance, not a comparison between conditions.
PValue	BH-corrected p-value for differential abundance
PClass	Sig or insignificant of the sequence

3.16.5 edgeR Plot

Scatter plot with logCPM on the x-axis and logFC on the y-axis. Significant sequences (below p-value cutoff) are colored differently from non-significant ones.

- Click **edgeR plot** (requires results in table).
- Enable **Adjust default edgeR plot** to customize the X-axis label (log CPM), Y-axis label (log fold change), plot title, and point colors for significant and non-significant sequences.

3.17 FASTAptamer-Cluster

3.17.1 Description

Group your counted FASTA sequences into families based on sequence similarity. FASTAptamer 3.0 uses a greedy Levenshtein Edit Distance (LED) algorithm — starting from the most abundant sequence as the first cluster seed, it pulls in all similar sequences within your chosen edit distance threshold. Any sequence that doesn't fit into a cluster is labeled NC (Not Clustered). This is the recommended first step for identifying dominant aptamer families in your selection.

3.17.2 Usage

- Click **Browse...** to upload a counted FASTA file.
- Set minimum number of reads to cluster, maximum LED, and maximum number of clusters to generate.
- Click **Start**. Results appear in the table on the right.
- Click **Download** to save the output as FASTA or CSV

3.17.3 Parameters

Parameter	Default	Range	Description
Min. number of reads to cluster	10	0–1,000	Sequences with fewer reads are excluded before clustering. Higher values shorten runtime.
Max. LED	7	1–20	Maximum edit distance between a sequence and a cluster seed for cluster membership. Use FASTAptamer-Distance to guide this threshold.
Max. number of clusters to generate	20	5–1,000	Caps the number of clusters produced. Remaining sequences are labeled NC.
Keep non-clustered sequences?	No	Yes / No	If Yes, NC-labeled sequences are retained in the output.
FASTA or CSV download?	FASTA	FASTA / CSV	Output format.

3.17.4 Output Table Columns

Column	Description
ID	Sequence identifier
Cluster	Assigned cluster number
Rank In Cluster	Rank within the cluster by read count
LED	Levenshtein edit distance from the cluster seed
Reads	Raw read count
Rank	Global rank across the population
RPU	Reads per unit
Sequence	Nucleotide or amino acid sequence

3.17.5 Extract Cluster

Select the target cluster and export the results as a CSV file containing only the filtered cluster.

3.18 FASTAptamer-Cluster Diversity

3.18.1 Description

Explore the diversity within each aptamer cluster. Upload a clustered FASTA file and FASTAptamer 3.0 calculates key statistics for each cluster — number of unique sequences, total reads, total RPU, and average

edit distance from the seed sequence. Results are displayed in a summary table alongside cluster metaplots and k-mer composition plots.

3.18.2 Usage

- Click **Browse...** to upload a clustered FASTA or previously generated diversity CSV file.
- If uploading a previously generated Cluster Diversity result file, click **Visualize Results** to load it directly into the table or plots without re-running.
- Click **Start** to compute statistics. Results appear in the table.
- Click **Download** to save the diversity CSV.

3.18.3 Output Table Columns

Column	Description
Cluster	Cluster number
Total Sequences	Number of unique sequences within the cluster
Total Reads	Sum of all read counts within the cluster
Total RPU	Sum of RPU values within the cluster
Average LED	Mean edit distance from the seed
SeedID	ID of the cluster's seed sequence

3.18.4 Cluster Metaplots

This plotting generates three stacked line charts per cluster: unique sequence count (top), total reads (middle), average LED (bottom).

- Click **Cluster Metaplots** after running analysis.
- Enable **Adjust default plots of cluster metadata** to customize the X-axis label, Y-axis 1 label and color (unique sequence count), Y-axis 2 label and color (total read count), Y-axis 3 label and color (average LED), and the plot title.

3.18.5 k-mer Plot (PCA / UMAP)

Visualize the sequence diversity within your selected clusters in 2D. Each sequence is represented by its k-mer composition — the frequency of all possible short sub-sequences of length k — which is then projected into two dimensions for easy comparison. Sequences from the same cluster that group closely together share similar sequence patterns, while spread-out clusters indicate greater internal diversity.

- Select clusters from the Available chip list.
- Choose k-mer size: 3, 4, or 5.
- Choose Plot type: PCA or UMAP.
- Click **k-mer** plot. Note: Characters outside [A, C, G, T, U] are converted to X before k-mer computation.
- Enable **Adjust default k-mer plot** to customize the X-axis label (first reduced dimension), Y-axis label (second reduced dimension), legend title (cluster color legend), plot title, and color palette (Accent, Dark2, Paired, Pastel1, Pastel2, Set1, Set2).

3.19 FASTAptamer-Cluster MSA (Multiple Sequence Alignment)

3.19.1 Description

Run Multiple Sequence Alignment (MSA) on all sequences within a selected cluster. Upload a clustered FASTA file, choose your target cluster, and FASTAptamer 3.0 outputs an aligned FASTA along with Shannon entropy and mutual information plots to reveal conserved and co-varying positions. Note: MSA is computationally intensive — runtime increases with cluster size. Please remain on the page until processing completes.

3.19.2 Usage

- Click **Browse...** to upload a clustered FASTA file.
- To load an existing MSA result directly, click **Visualize Results**.
- Select the cluster number from the dropdown.
- Choose Type of sequences: Nucleotide or AminoAcid.
- Choose download format: FASTA or CSV.
- Click **Start**. Aligned sequences appear in the table.
- Click **Download** to save the output.

3.19.3 Parameters

Parameter	Default	Description
Cluster	–	Cluster number to align (populated from uploaded file)
Type of sequences	Nucleotide	Nucleotide or AminoAcid — selects the appropriate substitution model
FASTA or CSV download?	FASTA	Output format for the aligned sequences

3.19.4 MSA Alignment Chart

Stacked bar chart showing nucleotide/amino acid composition at each alignment position.

- Click **MSA alignment chart** (requires completed run with data in table).

3.19.5 MSA Entropy Plot

Bar chart of Shannon entropy (in nats) per alignment position.

- Click **MSA entropy** after a successful run.
- Enable **Adjust default entropy plot** to customize the X-axis label (alignment position), Y-axis label (entropy in nats), legend title, plot title, and bar outline and fill colors.

3.19.6 MSA Mutual Information Plot

Heatmap of pairwise mutual information between all alignment positions, revealing co-varying positions.

- Click **MSA mutual information** after a successful run.
- Enable **Adjust default mutual information plot** to customize the X-axis label, Y-axis label (alignment positions), legend title (MI color scale), plot title, and fill color palette.

3.20 FASTAptamer-Cluster PHMM (Profile Hidden Markov Model)

3.20.1 Description

Generate new aptamer sequences that statistically resemble your enriched cluster. This module builds a Profile Hidden Markov Model (PHMM) from your cluster's MSA output, capturing the sequence preferences at every position. FASTAptamer 3.0 then simulates new sequences that match this profile — enabling in-silico exploration of your cluster's sequence space without additional wet-lab rounds. Outputs include the PHMM model file and a FASTA of simulated sequences.

3.20.2 Usage

1. Upload a cluster MSA FASTA file (output from the MSA tab).
2. Set the number of sequences to simulate and the target sequence length.
3. Click **Start**.
4. Click **Download PHMM** to save the model file or **Download sequences** to save the simulated FASTA.

3.20.3 Parameters

Parameter	Default	Range	Description
Simulate how many sequences?	100	10–1,000	Number of new sequences to generate from the PHMM
Length of simulated sequence (including gaps)	50	10–200	Target length of each simulated sequence, counting gap characters

3.21 FASTAptamer-Recluster

3.21.1 Description

Unify related aptamer families across multiple selection rounds. This module compares the seed sequences of clusters from separately clustered FASTA files using pairwise Levenshtein Edit Distance. Cluster families whose seeds fall within your chosen edit distance threshold are merged into a single super-cluster, making it easier to track dominant aptamer families across populations.

Two modes are available:

- **2 Populations** — cross-compare two populations (e.g., pre- and post-selection). Outputs a cluster-level summary table and a sequence-level table with per-sequence enrichment.
- **Multi-round (3 files)** — track cluster family convergence across three consecutive selection rounds. Outputs a single merged table with abundance and enrichment columns for all three rounds.

Avg RPU — mean RPU across all sequences assigned to the super-cluster in that population. Reflects overall cluster family abundance but is pulled down by low-count peripheral variants.

Seed RPU — RPU of the cluster seed (the most abundant sequence in the original cluster). More robust than Avg RPU because it represents the dominant variant only, unaffected by rare sequences at the cluster periphery.

Use Seed RPU to compare dominant aptamers between rounds. Use Avg RPU to capture the enrichment of the entire cluster family.

3.21.2 Usage-2 Populations Mode

- Select 2 Populations mode.
- Upload Cluster data 1 (first population clustered FASTA or CSV).
- Upload Cluster data 2 (second population clustered FASTA or CSV).
- Set the maximum LED threshold.
- Set the parameters used to define the Status column, including the status classification metric and the log₂E threshold.
- Click **Recluster**. The cluster-level summary table appears on the right; if sequence-level data is available, it appears below.
- Click **Download Summary** to save the cluster-level CSV.
- Click **Download Reclustered Sequences** to save the sequence-level CSV.

3.21.3 Usage-Multi-round Mode

- Select Multi-round (3 files) mode.
- For each round (Round 1, Round 2, Round 3), enter a Round label (e.g., LR8, LR10, LR15) and upload the corresponding clustered FASTA or CSV.
- Set the maximum LED threshold.
- Click **Recluster**. Results appear in the table.
- Click **Download Summary** to save the output.

3.21.4 Parameters

Parameter	Default	Range	Description
Max. LED	7	1–20	Maximum edit distance between seeds from different populations for them to be merged into the same super-cluster.
Status classification metric (two-populations only)	Avg RPU enrichment	Avg or Seed RPU enrichment	Choose which enrichment value is used to assign cluster status (enriched / depleted / inherited)
log2E threshold (two-populations only)	1	0.5-10	Sets the log2 fold-change cutoff for status assignment. Example: threshold = 1 means $\log_2 E \geq 1 \rightarrow$ Enriched (2× increase); $\log_2 E \leq -1 \rightarrow$ Depleted (2× decrease); between -1 and 1 $\rightarrow$ Inherited.
Round label (multi-round only)	R1, R2, R3	—	Display name for each round, used as column suffixes in the output (e.g., "LR8" $\rightarrow$ AvgRPU.LR8).

3.21.5 Output Table Columns — 2 Populations

Cluster-level summary:

Column	Description
Super-cluster	Merged cluster identifier
Seed	Seed sequence of the super-cluster
Size (Pop 1 / Pop 2)	Number of unique sequences in each population
Avg RPU (Pop 1 / Pop 2)	Mean RPU across all sequences in the super-cluster per population
Enrichment (Avg)	Avg RPU (Pop 2) / Avg RPU (Pop 1)
log2E (Avg)	$\log_2$ of the average RPU enrichment
Seed RPU (Pop 1 / Pop 2)	RPU of the cluster seed in each population
Enrichment (Seed)	Seed RPU (Pop 2) / Seed RPU (Pop 1)
log2E (Seed)	$\log_2$ of the seed RPU enrichment
Status	Whether the super-cluster was found in both populations or only one

Sequence-level data:

Column	Description
Sequence	Nucleotide or amino acid sequence
Super-cluster	Merged cluster assignment
Orig. Cluster (Pop 1 / Pop 2)	Original cluster number from each input file
Rank In Cluster	Rank within the super-cluster by read count
LED	Edit distance from the super-cluster seed
RPU (Pop 1 / Pop 2)	RPU of the sequence in each population
Enrichment	RPU (Pop 2) / RPU (Pop 1) per sequence
log2E	$\log_2$ per-sequence enrichment

3.21.6 Output Table Columns — Multi-round

Columns are named using the round labels you supplied (e.g., R1, R2, R3):

Column	Description
Super-cluster	Merged cluster identifier
Seed	Seed sequence of the super-cluster
AvgRPU.R1 / R2 / R3	Mean RPU across all super-cluster sequences per round
SeedRPU.R1 / R2 / R3	RPU of the cluster seed per round
E.R1.R2	AvgRPU.R2 / AvgRPU.R1 — average RPU enrichment from round 1 to round 2
E.R2.R3	AvgRPU.R3 / AvgRPU.R2 — average RPU enrichment from round 2 to round 3

SeedE.R1.R2	SeedRPU.R2 / SeedRPU.R1 — seed RPU enrichment from round 1 to round 2
SeedE.R2.R3	SeedRPU.R3 / SeedRPU.R2 — seed RPU enrichment from round 2 to round 3

3.21.7 LED Heatmap

Pairwise LED between all cluster seeds of population 1 (x-axis) vs. population 2 (y-axis).

- Click **LED Heatmap** (both files must be uploaded before running).
- Enable **Adjust default heatmap** to customize the X-axis label (population 1 clusters), Y-axis label (population 2 clusters), legend title (LED color scale), plot title, and color palette (Magma, Inferno, Plasma, Viridis, Cividis, Hot, Jet, Turbo).

3.21.8 Population Plot

Side-by-side bar chart comparing unique sequence counts per super-cluster between the two populations.

- Click **Population Plot** (requires recluster results in the table).
- Enable **Adjust default bar plot** to customize the X-axis label (cluster axis), Y-axis label (unique sequence count), legend title (population legend), plot title, and bar color for each population.

3.21.9 RPU Plot

Compares mean Avg RPU per super-cluster between the two populations.

- Click **RPU Plot**
- Enable **Adjust default bar plot** to customize the X-axis label (cluster axis), Y-axis label (unique sequence count), legend title (population legend), plot title, and bar color for each population.

3.21.10 Enrichment Plot

Bar chart of enrichment (Avg RPU Pop 2 / Pop 1) per super-cluster.

- Click **Enrichment Plot**.
- Enable **Adjust default enrichment plot** to customize the X-axis label (cluster axis), Y-axis label (enrichment), plot title, and bar fill and outline colors.

3.21.11 Trajectory Plot

Only available in multi-round mode. Line chart showing how Avg RPU evolves across the three rounds for each super-cluster.

- Click **Trajectory Plot** (requires recluster results in the table)
- Enable **Adjust default plot** to customize the Y-axis label (RPU) and plot title.

3.22 FASTAptamer-Pipeline

3.22.1 Description

Chain multiple FASTAptamer modules into a single automated workflow. Upload your starting file, add and configure your analysis steps in order, and run the entire pipeline with one click — each step's output is automatically passed as input to the next. Progress, input/output filenames, and any error messages are displayed in real time in the right panel. Available steps include Preprocess, Count, Translate, Motif Search, Motif Omit, Motif Discovery, Mutation Network, Sequence Distance, Cluster, Cluster MSA, Cluster PHMM, and Cluster Diversity. Note: please remain on this page while the pipeline is running. All temporary files are deleted when the session is closed or refreshed.

3.22.2 Usage

- Click **Browse...** to upload an initial FASTA or FASTQ file.
- Click **Add Step** to add operations to the pipeline. Each step appears as a card.
- For each step, select an operation from the dropdown and configure parameters.

- Reorder or remove steps as needed.
- Click **Run Pipeline**. Steps execute sequentially; the right panel shows real-time status.
- Click **Download** on any completed step to save its output.
- Click **Cancel** to stop the pipeline mid-run.

3.22.3 Available Operations and Parameters

Operation	Key Parameters	Valid Inputs
Preprocess	• Flanking 5' region • Flanking 3' region • Adapter matching error rate • Fixed bases to trim from 5' end • Fixed bases to trim from 3' end • Sequence length range (Min / Max) • Maximum allowed average error probability	FASTA / FASTQ
Count	Reverse complement, scaling factor	Preprocessed or raw FASTA/FASTQ
Translate	Open reading frame (1/2/3), merge identical (yes/no), genetic code	Counted FASTA
Motif Search	Motifs (comma-separated), highlight, partial match, motif type	Counted FASTA
Motif Omit	Motifs (comma-separated), partial match, motif type	Counted FASTA
Motif Discovery	Min reads, min/max motif length, alphabet (nucleotide/protein)	Counted FASTA
Mutation Network	Start sequence, end sequence, max edit distance	Counted FASTA
Sequence Distance	Query sequence	Counted FASTA
Cluster	Min reads, max LED, max clusters, keep NC (yes/no)	Counted FASTA
Cluster MSA	Cluster ID, sequence type (nucleotide/protein)	Clustered FASTA
Cluster PHMM	Number of sequences to simulate, target sequence length	Cluster MSA FASTA
Cluster Diversity	(no parameters)	Clustered FASTA

3.22.4 Step Ordering Rules

The pipeline enforces valid step sequences. Some operations are only valid following specific prior steps:

- **Count** must follow **Preprocess** (or be the first step if starting from a counted file is not intended).
- **Cluster** accepts output from **Count**, **Translate**, **Motif Search**, or **Motif Omit**.
- **Cluster MSA** and **Cluster Diversity** must follow **Cluster**.
- **Cluster PHMM** must follow **Cluster MSA**.

Invalid step combinations are flagged with a warning icon and an error message on the step card.

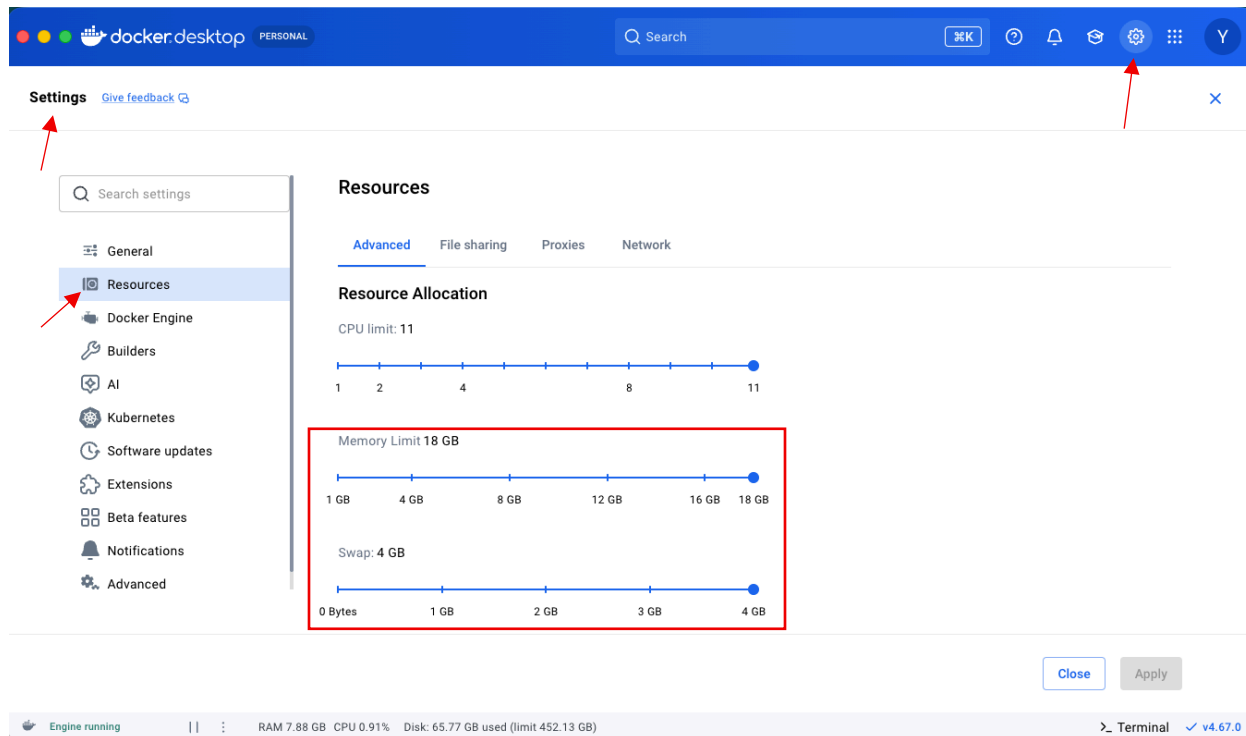
4 Known issues

4.1 SSE connection failure during Preprocess

This usually indicates that the application was terminated due to excessive memory usage. We can check your memory setting in Docker Setting-Resources. Please increase the memory allocation in Docker Desktop settings.

- **macOS:** Docker Desktop → Settings → Resources → Memory

- **Windows (WSL2 backend):** Run the provided setup-docker-resources.ps1 script from the repository (https://github.com/MU-DBL/fastaptamer3_python) to increase the memory available to Docker.
 - Open PowerShell in the **ps1 file** directory
 - Run: `.\setup-docker-resources.ps1`
 - Restart WSL: `wsl --shutdown`



4.2 File format compatibility issue

Although FASTAptamer 3.0 (FA3) Count files superficially resemble those produced by FASTAptamer 2.0 (FA2), the two formats are not interchangeable due to differences in metadata structure and parsing logic. Attempting to use FA2 Count files as direct input to FA3 downstream modules will result in compatibility errors. Users should therefore import original, uncounted sequence files and regenerate Count files natively within the FA3 environment.

External preprocessing platforms such as Galaxy are fully compatible with FA3 for quality control and adapter trimming; however, any FASTA or FASTQ files produced externally must pass through the FA3 Count module before use in downstream analysis.

The Count module serves as FA3's normalization gateway: it converts raw sequence data into a standardized internal format required uniformly by all downstream modules, ensuring consistent data representation throughout the analysis pipeline.

4.3 Window freeze and unresponsive issue

Users may experience an unresponsive browser window or temporarily freezing during certain analytical operations. This behavior usually indicates that computation is still running in the background, rather than an application error. Computationally intensive tasks, such as sequence counting, clustering, or differential enrichment analysis of large libraries, may require substantial processing time. During these operations, the

interface may become temporarily unresponsive while the job is running. Users should allow the process to be completed and should not refresh or close the browser window, as doing so may terminate the job.

If the browser remains unresponsive for an extended period or if jobs repeatedly fail to complete, performance may be improved by increasing the CPU and memory resources allocated to Docker. This is especially recommended when processing libraries with high sequence diversity, such as early-round libraries, or when running multiple analyses simultaneously. Instructions for increasing Docker CPU and memory allocation are provided in Issue 4.1.

The memory and runtime requirements of the count module depend primarily on library diversity, especially the number of unique sequences, rather than file size alone. Highly diverse libraries require more memory and longer processing times than enriched libraries of similar file size.

Estimated RAM and runtime requirements for the count module

Unique sequences	Typical library type	Estimated RAM	Estimated runtime
< 100K	Late or enriched rounds	< 500 MB	< 1 min
100K–1M	Mid-selection rounds	500 MB–1 GB	1–3 min
1M–5M	Early rounds	1–3 GB	3–8 min
5M–10M	Naïve or R0 libraries	3–6 GB	8–15 min
> 10M	Very high-diversity libraries	6–12 GB	15–30 min

Recommended Docker resource allocation

Workload	Recommended RAM	Recommended CPUs
File < 1 GB, enriched library	6 GB	2
File 1–3 GB, mid-diversity library	10 GB	4
File 3–5 GB, high-diversity library	16 GB	6
File > 5 GB or naïve library	32 GB	8